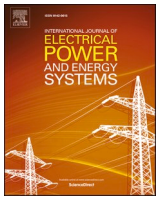
Electrical Power and Energy Systems 150 (2023) 109093



Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes





DC fault detection in hybrid MTDC systems using transient average of DC reactor voltage

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ARTICLE INFO

Keywords:
Short-circuit current calculation
Hybrid MTDC system
Fault detection
Transient average voltage

ABSTRACT

The existence of various types of converters makes the fault characteristics of the hybrid multi-terminal DC (MTDC) system more complicated than the voltage source converter-based (VSC-based) MTDC system. A reduced line commutated converter (LCC) model for DC fault analysis is first proposed in this paper, where the LCC-based rectifier with control effects is regarded as an equivalent RLC circuit. By neglecting the resistive and capacitive components in the high-frequency domain, the original network analysis is significantly simplified to derive concise time-domain expressions of the initial DC fault current. On top of this, a single-terminal DC fault detection scheme using the transient average of DC reactor voltage is designed for the hybrid MTDC system. The thresholds for detecting faults on the LCC-connected and VSC-connected lines are calculated based on the derived fault current expressions, respectively. Simulation studies based on PSCAD/EMTDC have validated the proposed fault analysis model is accurate within the initial period after fault, while the proposed fault detection scheme for hybrid MTDC systems is fast, sensitive, and tolerant to high fault resistances.

1. Introduction

Traditional LCC-based high voltage direct current (HVDC) systems have prevailed in long-distance power transmission because of their high capacity and low cost [1]. However, line commutated converter-based (LCC-based) inverters are vulnerable to commutation failures if the voltage of the point of common coupling is relatively low [2]. In recent years, there has been substantial development in voltage source converters (VSCs) [3,4], especially the modular multi-level converter (MMC) [5]. Despite the higher cost, VSCs have various advantages over LCCs, including commutation failure elimination, needlessness of AC system supports, flexible reactive power control, etc. Incorporating the advantages of LCCs and VSCs, the hybrid multi-terminal DC (MTDC) system is promising for renewable energy integrations and asynchronous interconnections of multi-regional AC grids in the future [6]. Considering hybrid MTDC systems consist of different types of converters with various dynamic properties, it is pending to identify fault characteristics and design reliable DC fault detection schemes for hybrid LCC-VSC MTDC systems, imminently.

1.1. Related works

Calculation of short-circuit current is essential to select power devices and design protection schemes. Electromagnetic transient simulation (EMT) is one of the most common tools to derive system fault current dynamics. The system is modelled as a series of differential–algebraic equations (DAEs), and the dynamics of concerned electrical quantities can be obtained by solving the DAEs numerically. The detailed simulation model of the LCC-MMC three-terminal HVDC transmission system (China Wudongde HVDC project) is reported [7]. The system's fault characteristics can be assessed based on the simulation results. In [8], a dynamic phasor model of a two-terminal LCC-MMC system is proposed to improve the computation efficiency. Unfortunately, fault analysis based on simulation only applies for several specified cases, rather than the generic ones on revealing the short-circuit current characteristics.

To tackle this issue, great efforts have been made to derive the closed-form expressions of DC fault currents. The two-level VSC is simplified to a DC-link capacitor before blocking insulated gate bipolar transistors (IGBTs) [9]. Analogously, MMCs are proven to be equivalent to a series-connected RLC circuit within the first instants after DC fault

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https://doi.org/10.1016/j.ijepes.2023.109093
Received 19 September 2022; Received in revised form 26 December 2022; Accepted 9 March 2023
Available online 20 March 2023
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[10]. With these reduced models of converters, the closed-form solution of fault current in VSC-MTDC systems is obtained based on the linear circuit analysis [11]. The simplified analytical expression of the short-circuit current is further derived in [12]. So far, investigations on analytical fault analysis of hybrid MTDC systems are in their infancy. Due to the combined effects of LCCs and VSCs, fault characteristics of hybrid MTDC systems differ from VSC-MTDC systems, which demand further exploration.

Fast fault isolation is critical to ensure continuous power transmission in MTDC systems. Considering communication delay of pilot protection will slow down the fault discrimination, the single-terminal protection can be more suitable as the prime protection for MTDC grids [13]. Single-terminal travelling wave (TW) protection has been widely applied in industries for its ultra-fast speed and high sensitivity. Protection schemes based on the rate of change of current and voltage (ROCOC and ROCOV) are proposed in [14,15], respectively. Leveraging the difference in arrival time between the earth mode and line mode of the first TW, a single-terminal protection algorithm for long-distance overhead lines in MTDC systems is proposed [16]. However, TW-based methods require relatively high sampling frequency (several hundreds of kHz or even 1 MHz) and are of low tolerance to high fault resistances. To improve the protection performance, the ROCOV and the transient voltage across the DC reactor are exploited and proven to be sensitive to DC faults [17,18]. By using the line boundary consisting of the DC reactors and the capacitor of VSCs, a robust fault detection scheme for MTDC systems based on transient average current is proposed in [19].

Various types of converters make the line boundary characteristics of hybrid MTDC systems different from VSC-MTDC systems, which impacts the adaptability of single-terminal protection schemes designed based on VSC-MTDC systems. To date, however, there are only a few works that concentrate on the protection of hybrid MTDC systems. The mathematical expression of the first TW is derived in [20], and a convolution-power-based fault detection scheme of LCC-MMC MTDC systems is further proposed. The first peak time of the TW of line-mode voltage is utilized to improve the sensitivity of the fault detection scheme in [21]. The phase of voltage TW is adopted as a fault detection indicator in [22] to enhance the discernment of backward external faults. Nevertheless, existing methods rely on the exact acquisition of the first TW, which not only require comparatively high sampling frequency but also are inherently vulnerable to error data or data missing.

1.2. Main contribution

The major contributions of this paper include the following two aspects.

- Concise time-domain expressions of short-circuit current in hybrid LCC-VSC MTDC systems are first deduced. The original LCC model is simplified so that LCC incorporating the control effects is equivalent to a series-connected RLC circuit. By contrast, the control effects of VSC are overlooked during the first instants after the DC fault. Within this initial period, concise mathematical expressions of fault current are obtained by reserving only the high-frequency components of the Laplace circuit of the fault component network.
- A single-terminal DC fault detection scheme using the transient average of DC reactor voltage is designed for hybrid LCC-VSC MTDC systems. Instead of leveraging the first TW, the transient average of DC reactor voltage persists on a longer time scale. Compared with existing methods, the proposed method distinguishes itself for relatively low sampling frequency requirement and robustness against noise corruption.

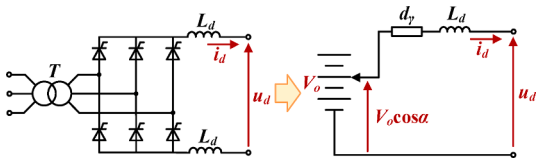


Fig. 1. Basic structure and equivalent circuit of the LCC-based rectifier.

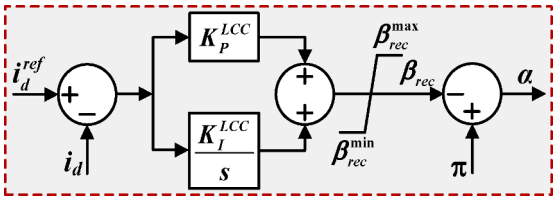


Fig. 2. Diagram of the constant current control of LCC.

2. Modelling of converters

2.1. LCC model

In hybrid HVDC systems, LCCs are normally operated as rectifiers, while MMCs are operated as inverters to avoid commutation failure and enhance the stability of receiving-end AC grids. Consequently, only the LCC-based rectifier is taken into account in this paper. The basic structure and equivalent circuit of the LCC-based rectifier are shown in Fig. 1, and the dynamics of LCC are written as follows.

$u_d = V_o\cos\alpha - d_\gamma i_d - L_d\frac{di_d}{dt}$.where L_d is the inductance of the smoothing reactor. d_γ is the equivalent commutation resistance, and $d_\gamma = 3\omega_s L_\gamma/\pi$, in which ω_s is the grid frequency, and L_γ is the equivalent inductance between LCC and its connected AC source. u_d and i_d are the DC-link voltage and current, respectively. V_o is the ideal no-load voltage of LCC and$V_o = 3\sqrt{2}E/\pi$, in which E is the root-mean-square (RMS) line voltage of the AC source. α denotes the firing angle of LCC.

Fig. 2 depicts the constant current control of LCC. The proportional-integral controller will restrict the current surge during DC faults, which is written as follows.

$$\alpha = \pi - K_P^{LCC}\left(i_d^{ref} - i_d\right) - K_I^{LCC}\int\left(i_d^{ref} - i_d\right)dt$$

where *iref d* is the reference DC-link current. *KLCC P* and *KLCC I* are the proportional and integral gains of the constant current control of LCC, respectively.

Because of the nonlinear term of cosα in, it is difficult to directly deduce analytical time-domain expressions of DC fault current as per and. To simplify the analysis, a linear approximation of cosα is constructed based on Taylor expansion.

Taylor expansion of cosα at the initial steady-state point is expressed as,

$$\begin{aligned}\cos\alpha &\approx \cos\alpha_0 - \sin\alpha_0(\alpha-\alpha_0) - \frac{\cos\alpha_0}{2}(\alpha-\alpha_0)^2 \\ &= \cos\alpha_0 - \left(\sin\alpha_0 + \frac{\cos\alpha_0(\alpha-\alpha_0)}{2}\right)(\alpha-\alpha_0)\end{aligned}$$

where α_0 is the initial firing angle, which is normally configured in the range of [π/36, π/12] for LCC-based rectifiers to enhance the power factor of the transmission and reserve certain regulation capability of the DC-link voltage. The small value of α_0 makes cosα_0 become much larger than sinα_0 such that the first-order Taylor expansion is not accurate enough to approximate cosα. Hence, the second-order Taylor expansion is utilized in.

Denote α_1 as the post-fault steady-state value of the firing angle. In the range of [α_0, α_1], the average value can be used to approximate α.

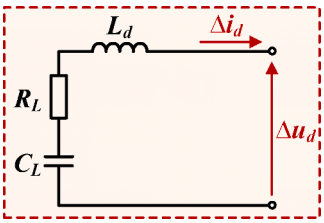


Fig. 3. Proposed LCC model for DC fault analysis.

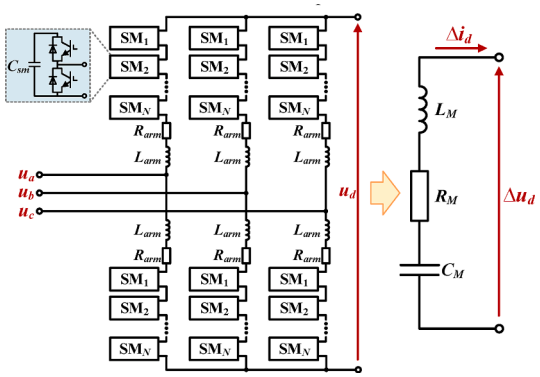


Fig. 4. Basic structure and the DC fault model of MMC.

Taking the relation of $\alpha \approx (\alpha_0 + \alpha_1)/2$ into yields,

$$\cos\alpha \approx \cos\alpha_0 - \left(\sin\alpha_0 + \frac{\cos\alpha_0((\alpha_0+\alpha_1)/2-\alpha_0)}{2}\right)(\alpha-\alpha_0)$$
$$= \cos\alpha_0 - \left(\sin\alpha_0 + \frac{\cos\alpha_0(\alpha_1-\alpha_0)}{4}\right)(\alpha-\alpha_0) = \cos\alpha_0 - k_\alpha(\alpha-\alpha_0)$$

provides a linear approximation of cosα. After the DC fault, the firing angle of LCC fast surges to around $\pi/2$ by the constant current controller. Consequently, α_1 is chosen as $\pi/2$ in this paper.

Substituting into and using Laplace transformation, it provides that,

$$\Delta u_d = -k_\alpha V_o \Delta\alpha - (d_\gamma + sL_d)\Delta i_d$$

where Δ denotes a generic deviation from the equilibrium.

Combining with and, the following equation holds,

$$\frac{\Delta i_d}{\Delta u_d} = -\frac{1}{sL_d + (d_\gamma + k_\alpha K_P^{LCC} V_o) + \frac{k_\alpha K_I^{LCC} V_o}{s}}$$

indicates that the terminal characteristics of LCC can be represented by an equivalent series-connected RLC circuit, as shown in Fig. 3. R_L and C_L are individually the equivalent resistance and capacitance of the LCC-based rectifier considering the control effects, expressed as,

$$\begin{cases} R_L = d_\gamma + k_\alpha K_P^{LCC} V_o \\ C_L = \dfrac{1}{k_\alpha K_I^{LCC} V_o} \end{cases}$$

2.2. MMC model

The basic structure of the three-phase MMC is shown in Fig. 4. Each phase has the upper and lower arms consisting of N series-connected submodules (SMs). During normal operation, each SM will be either on-state or off-state, with the terminal voltage equal to the capacitor voltage or zero, respectively. The AC-side voltage of MMC denoted by $u_{a,b,c}$ and the DC-link voltage denoted by u_d can be regulated effectively by tuning the number of on-state SMs in the upper and low arms. The capacitance of each SM is denoted by C_{sm}. R_{arm} and L_{arm} are the equivalent arm resistance and inductance, respectively.

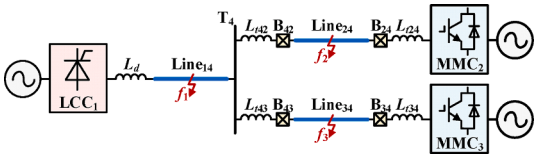


Fig. 5. Schematic of a typical three-terminal hybrid HVDC system.

After a DC fault, SM capacitors discharge into the fault point, causing a large DC current surge flowing through the transmission line and arms of the MMC. Within the initial fault period, around 5 to 10 ms, the drop of capacitor voltage and the change of AC feed current remain rather small [23,24]. Afterwards, IGBTs in the converter will be blocked to avoid overcurrent damaging the switches.

In hybrid MTDC systems, DC faults must be isolated before the block of IGBTs to avoid interrupting the power transmission of the whole DC grid. Consequently, fault current characteristics at the capacitor discharging stage will provide more important guiding significance to system protection design. In this stage, the control effects of MMC are negligible, and the equivalent circuit of MMC is a series-connected RLC circuit shown in Fig. 4 [10,11]. The equivalent resistance, inductance, and capacitance of MMC are merely determined by its physical parameters, expressed as,

$$\begin{cases} R_M = 2\left(R_{arm} + \sum R_{on}\right)/3 \\ L_M = 2L_{arm}/3 \\ C_M = 6C_{sm}/N \end{cases}$$

where ΣR_{on} is the sum of on-state resistance of diodes or IGBTs per arm.

Remark. This paper mainly focuses on the fault characteristics of hybrid MTDC systems within the initial fault period. In this stage, the constant current control of the LCC-based rectifier activates to suppress the overcurrent. In contrast, the fault current of the MMC is provided by the natural responses of the discharging capacitor, which is little influenced by the control effects. Therefore, only the control loop of the LCC is considered in the initial fault analysis.

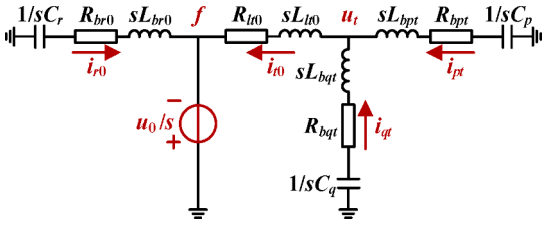


Fig. 6. Laplace circuit of the fault component network of the studied system under a metallic fault.

3. Short-circuit current calculation of hybrid MTDC system

3.1. Description of the studied system.

Fig. 5 shows the outline of a typical three-terminal hybrid HVDC transmission system. One LCC-based rectifier and two MMC-based inverters are connected via DC transmission lines. To limit the capacitor discharging current of MMCs after DC faults, supplementary DC reactors of several hundred mH are usually installed at the terminal of transmission lines. L_{tmn} denotes the inductance of the DC reactor on Line *mn* and nearby side *m*. B_{mn} represents the DC circuit breaker (DCCB) near the side *m* of Line *mn*. The implementation of DCCBs enables fast fault isolation in the multi-terminal HVDC system.

3.2. Fault current calculation under metallic DC faults

In DC grids, DC faults can be classified into the pole-to-pole (P2P) short-circuit fault, the pole-to-ground (P2G) short-circuit fault, and the breaking fault of DC lines. This paper mainly concentrates on the relatively more dangerous P2P short-circuit fault. As the hybrid bipolar HVDC system generally adopts the direct grounding scheme to avoid overvoltage under P2G faults, the fault characteristics under P2P and P2G faults are similar.

To simplify the analysis, transmission lines are represented by the classic RL model. Accordingly, the generic Laplace fault component network of the studied system incorporating a metallic DC fault is shown in Fig. 6. The fault point is marked as 0, and the pre-fault voltage of the fault point is u_0. The DC bus in the "T" area is denoted by the subscript *t*. Without loss of generality, the fault line is marked as Line *rt*, while the healthy lines are marked as Line *pt* and Line *qt*. The equivalent resistance and inductance of Line *mn* are marked as R_{lmn} and L_{lmn}, respectively. R_{bmn} and L_{bmn} denote the branch resistance and inductance of Line *mn* containing the transmission line, its connected converter and DC reactors, respectively. Taking the example of a DC fault occurring on Line 14, let $r = 1$, $p = 2$, $q = 3$, and the branch parameters can be obtained as follows.

$$\begin{cases} R_{br0} = R_{l10} + R_L, L_{br0} = L_{l10} + L_d, C_r = C_L \\ R_{bpt} = R_{l24} + R_{M2}, L_{bpt} = L_{l24} + L_{t24} + L_{t42} + L_{M2}, C_p = C_{M2} \\ R_{bqt} = R_{l34} + R_{M3}, L_{bqt} = L_{l34} + L_{t34} + L_{t43} + L_{M3}, C_q = C_{M3} \end{cases}$$

The case of DC fault on Line 24 or Line 34 can be represented similarly.

The fault line current i_{r0} can be directly calculated based on the left-hand-side (LHS) circuit of the fault point,

$$i_{r0} = \frac{u_0}{s(sL_{br0} + R_{br0} + 1/sC_r)}$$

To solve the right-hand-side (RHS) circuit of the fault point, the branch impedances of the healthy lines are first defined as,

$$\begin{cases} Z_{bpt} = sL_{bpt} + R_{bpt} + 1/sC_p \\ Z_{bqt} = sL_{bqt} + R_{bqt} + 1/sC_q \end{cases}$$

By Kirchhoff voltage law (KVL) and Kirchhoff current law (KCL), the following equations are obtained based on the RHS circuit of the fault point,

$$-u_t = Z_{bpt} i_{pt}$$

$$-u_t = Z_{bqt} i_{qt}$$

$$u_0/s + u_t = (sL_{lt0} + R_{lt0}) i_{t0}$$

$$i_{t0} = i_{pt} + i_{qt}$$

Combining (11), (12), (13) and (14), it yields that,

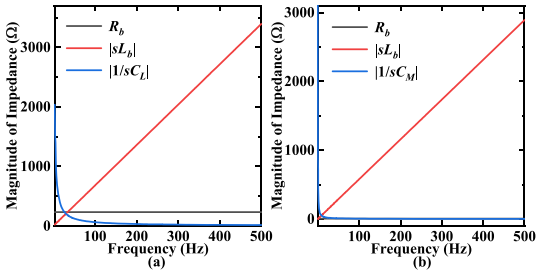


Fig. 7. Amplitude frequency characteristics of a 100 km DC transmission line connecting to an LCC and an MMC, respectively.

$$\begin{cases} i_{pt} = \dfrac{u_0}{s\left(Z_{bpt} + \left(Z_{bpt}/Z_{bqt} + 1\right)(sL_{lt0} + R_{lt0})\right)} \\ i_{qt} = \dfrac{u_0}{s\left(Z_{bqt} + \left(Z_{bqt}/Z_{bpt} + 1\right)(sL_{lt0} + R_{lt0})\right)} \end{cases}$$

An obvious way to deduce the time-domain expressions of short-circuit current is to calculate the inverse Laplace transformation of and. However, since the structures of the branch impedances Z_{bpt} and Z_{bqt} are relatively complicated, it is rather difficult to apply inverse Laplace transformation on directly. In fact, it is inaccurate and unnecessary to deduce full time domain expressions of fault current via inverse Laplace transformation as the effective period of the reduced MMC model does not exceed the capacitor discharging stage. Note that the high-frequency components of electrical quantities of the Laplace fault circuit dominate with the first instants after fault [12]. Hence, the initial DC fault analysis can be simplified by reserving the high-frequency components of the original Laplace fault circuit.

Normally, the fault characteristics within 5 ms after fault is more concerned for DC fault detection. Thus, the corresponding frequency cannot be less than 200 Hz. In the high-frequency domain, the following inequalities hold with typical parameters of HVDC projects.

$$\begin{cases} |sL_{bpt}| \gg R_{bpt} \gg |1/sC_p| \\ |sL_{bqt}| \gg R_{bqt} \gg |1/sC_q| \end{cases}$$

indicates the branch inductances affect the system dynamics more than the branch resistances and the converter capacitances within the initial fault period. Fig. 7 (a) and (b) illustrate the amplitude-frequency characteristics of a 100 km DC transmission line connecting to an LCC and an MMC with typical engineering parameters, respectively. It is observed that the inequalities in can be satisfied for both the LCC-connected and MMC-connected branches in the high-frequency domain.

Combining and, the ratio between Z_{bpt} and Z_{bqt} is reduced in the high-

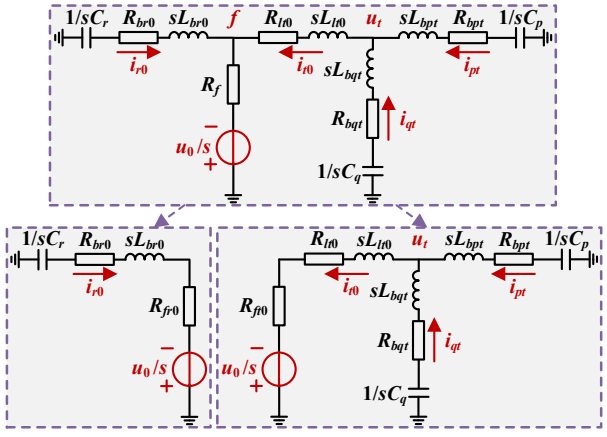


Fig. 8. Laplace fault component circuit and its decoupled circuit for fault analysis with fault resistance.

frequency domain as follows.

$$\frac{Z_{bpt}}{Z_{bqt}}=\frac{sL_{bpt}+R_{bpt}+1/sC_p}{sL_{bqt}+R_{bqt}+1/sC_q}\approx\frac{L_{bpt}}{L_{bqt}}$$

Substitution of into it provides that,

$$\begin{cases} i_{pt}=\dfrac{u_0}{s\left(Z_{bpt}+\left(L_{bpt}/L_{bqt}+1\right)\left(sL_{lt0}+R_{lt0}\right)\right)} \\ i_{qt}=\dfrac{u_0}{s\left(Z_{bqt}+\left(L_{bqt}/L_{bpt}+1\right)\left(sL_{lt0}+R_{lt0}\right)\right)} \end{cases}$$

By further neglecting the converter capacitances in the high-frequency domain, the reduced expressions of fault component current of the fault and healthy lines (taking the example of Line *pt*) are obtained based on and.

$$\begin{cases} i_{r0}=\dfrac{u_0}{s(sL_{br0}+R_{br0})} \\ i_{pt}=\dfrac{u_0}{s\left(sL_{bpt}+R_{bpt}+\left(L_{bpt}/L_{bqt}+1\right)\left(sL_{lt0}+R_{lt0}\right)\right)} \end{cases}$$

The time-domain expressions of the fault and healthy lines current are calculated by applying inverse Laplace transformation on.

$$\begin{cases} i_{r0}=\dfrac{u_0}{R_{br0}}\left(1-\mathrm{e}^{-\frac{t}{\tau_r}}\right), \qquad \tau_r=\dfrac{L_{br0}}{R_{br0}} \\ i_{pt}=\dfrac{u_0}{R_{bpt}+\left(L_{bpt}/L_{bqt}+1\right)R_{lt0}}\left(1-\mathrm{e}^{-\frac{t}{\tau_p}}\right), \tau_p=\dfrac{L_{bpt}+\left(L_{bpt}/L_{bqt}+1\right)L_{lt0}}{R_{bpt}+\left(L_{bpt}/L_{bqt}+1\right)R_{lt0}} \end{cases}$$

3.3. Fault current calculation with fault resistance

The Laplace fault component circuit of the studied system incorporating a DC fault with fault resistance of R_f is shown in Fig. 8. The original circuit can be decoupled into two independent parts divided from the fault point by introducing the equivalent branch fault resistances R_{fr0} and R_{ft0} as follows.

$$\begin{cases} R_{fr0}=\dfrac{i_{r0}+i_{t0}}{i_{r0}}R_f=\left(\dfrac{1}{k_{rt}}+1\right)R_f \\ R_{ft0}=\dfrac{i_{r0}+i_{t0}}{i_{t0}}R_f=(k_{rt}+1)R_f \end{cases}$$

where k_{rt} denotes the ratio between i_{r0} and i_{t0}, i.e., $k_{rt}=i_{r0}/\,i_{t0}$. By KVL and KCL, i_{r0} and i_{t0} are calculated as,

$$\begin{cases} i_{r0}=\dfrac{u_0-R_f(i_{r0}+i_{t0})}{s(sL_{br0}+R_{br0}+1/sC_r)} \\ i_{t0}=\dfrac{u_0-R_f(i_{r0}+i_{t0})}{s\left(sL_{lt0}+R_{lt0}+\left(Z_{bpt}//Z_{bqt}\right)\right)} \end{cases}$$

Based on, k_{rt} can be expressed as follows.

$$k_{rt}=\frac{i_{r0}}{i_{t0}}=\frac{sL_{lt0}+R_{lt0}+\left(Z_{bpt}//Z_{bqt}\right)}{sL_{br0}+R_{br0}+1/sC_r}$$

Combining and, k_{rt} can be reduced by neglecting the resistive and capacitive components in the high-frequency region as follows.

$$k_{rt}\approx\frac{sL_{lt0}+\left(sL_{bpt}//sL_{bqt}\right)}{sL_{br0}}=\frac{L_{lt0}}{L_{br0}}+\frac{L_{bpt}L_{bqt}}{L_{br0}\left(L_{bpt}+L_{bqt}\right)}$$

It is simple to solve the LHS circuit shown in Fig. 8. The RHS circuit is similar to that shown in Fig. 6, which can be solved analogously. The expressions of the fault and healthy lines current are written as follows.

$$\begin{cases} i_{r0}=\dfrac{u_0}{s\left(sL_{br0}+R_{br0}+(1/k_{rt}\ +\ 1)R_f\right)} \\ i_{pt}=\dfrac{u_0}{s\left(sL_{bpt}+R_{bpt}+\left(L_{bpt}/L_{bqt}+1\right)\left(sL_{lt0}+R_{lt0}+(k_{rt}+1)R_f\right)\right)} \end{cases}$$

In, the factor k_{rt} is constant for a given DC fault, which can be calculated based on. By applying inverse Laplace transformation on, it provides that,

$$\begin{cases} i_{r0}=\dfrac{u_0}{R_{br0}+(1/k_{rt}+1)R_f}\left(1-\mathrm{e}^{-\frac{t}{\tau_r}}\right) \\ i_{pt}=\dfrac{u_0}{R_{bpt}+\left(L_{bpt}/L_{bqt}+1\right)\left(R_{lt0}+(k_{rt}+1)R_f\right)}\left(1-\mathrm{e}^{-\frac{t}{\tau_p}}\right) \end{cases}$$

where,

$$\begin{cases} \tau_r=\dfrac{L_{br0}}{R_{br0}+(1/k_{rt}+1)R_f} \\ \tau_p=\dfrac{L_{bpt}+\left(L_{bpt}/L_{bqt}+1\right)L_{lt0}}{R_{bpt}+\left(L_{bpt}/L_{bqt}+1\right)\left(R_{lt0}+(k_{rt}+1)R_f\right)} \end{cases}$$

and serve as the analytical solution of initial fault current calculation in hybrid MTDC systems, which are the theoretical foundation for the following protection design.

4. Fault detection scheme

4.1. Transient average of DC reactor voltage

It has been shown that the DC reactor voltage is a sensitive indicator for fault detection [18]. Based on, the voltage across DC reactor L_{tmn} marked as u_{tmn} can be calculated as,

$$u_{tmn}=L_{tmn}\frac{di}{dt}=\frac{L_{tmn}u_0}{L_b}\mathrm{e}^{-\frac{t}{\tau}}$$

According to, small fluctuations of DC-link current may induce apparent changes in DC reactor voltage. In order to diminish the impacts of DC-link current fluctuations and noise corruption, the transient average of DC reactor voltage is utilized as a more robust and steadier indicator. Its definition is the averaged value of DC reactor voltage over a specified time period T.

$$\overline{u}(t)=\frac{1}{T}\int_{v=t-T}^{t}u(v)dv$$

Based on, it is simple to calculate the transient average of DC reactor voltage. The prominent advantages of the transient average of DC reactor voltage mainly include simple measurement, low complexity, robustness against noise, etc.

4.2. Threshold determination

To ensure the selectivity of the protection, the threshold value should be set higher than the maximum transient average of the DC reactor voltage under the severest external fault so that the protection will not mal-operate under external faults. Considering the change of DC reactor voltage can be obtained based on the fault analysis, the threshold determination is independent of numerical simulations.

As depicted in Fig. 5, once a fault happens on Line 14 (f_1), the power transmission of the entire HVDC system has to be interrupted, while B_{42} and B_{43} must fast open to isolate the DC fault. By contrast, when a fault occurs on Line 24 (f_2) or Line 34 (f_3), the DCCBs installed on the fault line must be activated so that the rest HVDC system can keep itself in operation. Thereby, the LHS DCCBs (B_{42} and B_{43}) may operate under both-side DC faults, while the RHS DCCBs (B_{24} and B_{34}) should be activated under a fault occurring on an MMC-connected line. In view of the differences between LHS and RHS DCCBs, the threshold determination of their corresponding protection units will be separately discussed.

For the RHS DCCBs, taking B_{24} for example, the severest external fault is a metallic fault occurring between L_{t43} and Line 34. The DC reactor voltage in this situation can be calculated based on and.

Table 1
Main parameters of the test hybrid MTDC system.

Symbol	Item	Value
E	RMS line voltage of the LCC-connected AC system	330 kV
i_d^{ref}	Reference value of the constant current control of LCC	1 kA
(K_P^{LCC}, K_I^{LCC})	Proportional and integral factors of the constant current control of LCC	(1, 100)
L_d	Inductance of smoothing reactor	400 mH
d_γ	Equivalent commutation resistance of LCC	36 Ω
N	Number of SMs per arm of MMC	200
C_{sm}	Capacitance of SM capacitor of MMC	12 mF
R_{on}	On-state resistance of one IGBT/diode	5.445 mΩ
R_{arm}	Arm resistance of MMC	0 Ω
L_{arm}	Arm inductance of MMC	60 mH
L_t	Inductance of DC reactors	150 mH
R_0	Per-kilometer resistance of DC line	0.015 Ω
L_0	Per-kilometer inductance of DC line	1.4 mH
l_{14}, l_{24}, l_{34}	Length of Line 14, Line 24 and Line 34	50, 100, 100 km
f_s	Sampling frequency	10 kHz
T	Time window for averaging	2 ms
$u_t^{set,I}$, $u_t^{set,II}$	Threshold value for forward and backward faults	51.42, –23.21 kV

$$u_{t24} = \frac{L_{t24}u_0}{L_{b24} + (L_{b24}/L_{b14} + 1)L_{t43}} e^{-\frac{R_{b24}t}{L_{b24}+(L_{b24}/L_{b14}+1)L_{t43}}}$$

Based on, the DC reactor voltage exponentially increases to its maximum value, so the transient average voltage cannot exceed this value. Therefore, the maximum transient average of DC reactor voltage can be calculated based on and as follows.

$$\overline{u_{t24}}(t) = \frac{1}{T}\int_{\upsilon=t-T}^{t} u_{t24}(\upsilon)d\upsilon \leqslant \frac{L_{t24}u_0}{L_{b24} + (L_{b24}/L_{b14} + 1)L_{t43}} = \max\{\overline{u_{t24}}(t)\}$$

As stated previously, the threshold value should be higher than the maximum transient average voltage. By introducing a reliability factor of more than 1, the threshold value can be calculated as,

$$u_{t24}^{set,I} = K_{rel}\max\{\overline{u_{t24}}(t)\} = \frac{K_{rel}L_{t24}u_n}{L_{b24} + (L_{b24}/L_{b14} + 1)L_{t43}}$$

where K_{rel} is the reliability factor with consideration of the model error, and it is set at 1.3 in this paper. u_n is the nominal DC-link voltage. Considering the DC-link voltage slightly fluctuates around its nominal value during normal operation, u_0 is substituted by u_n in. The fault detection criterion can be expressed as,

$$\overline{u_{t24}}(t) > u_{t24}^{set,\ I}$$

As for the LHS DCCBs, they must isolate both-side internal DC faults. For simplicity, faults occurring on the MMC-connected line are defined as forward faults, while faults occurring on the LCC-connected line are defined as backward faults for LHS DCCBs. Accordingly, there are two thresholds for the LHS DCCBs. The threshold for detecting forward faults can be determined analogously based on. As for the threshold for backward faults, the severest external backward fault is a metallic fault occurring between the LCC-based converter and the smoothing reactor L_d. Under this fault, the voltage over L_{t42} is obtained based on and,

$$u_{t42} = \frac{L_{t42}u_0}{L_{b24} + (L_{b24}/L_{b34} + 1)L_{b14}} e^{-\frac{R_{l14}t}{L_{b24}+(L_{b24}/L_{b34}+1)L_{b14}}}$$

Based on, the threshold for backward faults can be set as,

$$u_{t42}^{set,\ II} = K_{rel}\min\{\overline{u_{t42}}\} = -\frac{K_{rel}L_{t42}u_n}{L_{b24} + (L_{b24}/L_{b34} + 1)L_{b14}}$$

where the minus sign denotes the backward fault. Accordingly, the operating criterion of B_{42} is expressed as,

$$\overline{u_{t42}}(t) > u_{t42}^{set,\ I} \quad \text{or} \quad \overline{u_{t42}}(t) < u_{t42}^{set,\ II}$$

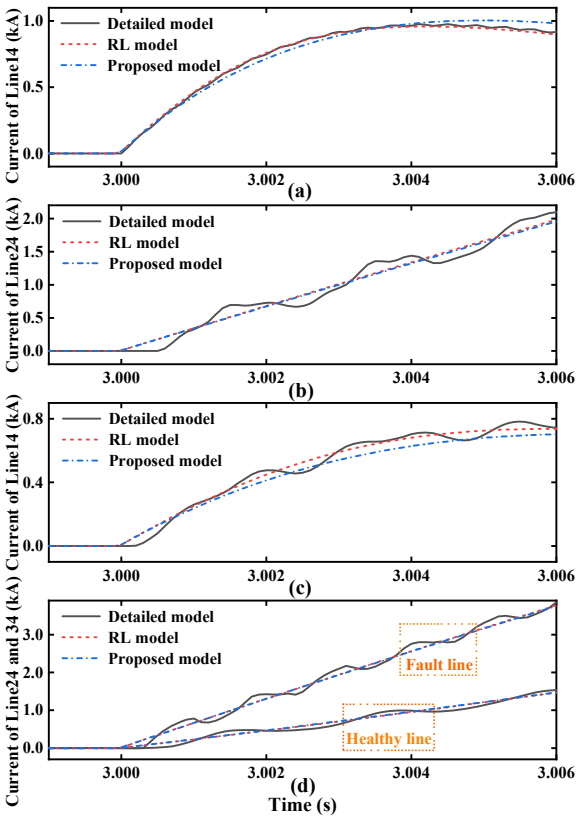


Fig. 9. (a) fault component current of Line 14 when f_1 occurs; (b) fault component current of Line 24 when f_1 occurs; (c) fault component current of Line 14 when f_2 occurs; (d) fault component current of Line 24 and Line 34 when f_2 occurs.

Based on and, the threshold value of the fault detection can be determined via simple calculation rather than numerical simulations.

4.3. Summary of the proposed method

The main procedure of the proposed method is stated as follows. Firstly, fault analysis of the hybrid MTDC system under study should be conducted, yielding the analytical expression of the fault current within the initial fault period as expressed by. Secondly, the threshold value of the proposed protection using the transient average voltage is determined based on the fault analysis. By considering the severest external fault, the thresholds for detecting forward and backward faults are calculated as per and, respectively. Thirdly, the transient average voltage of the DC reactor is calculated in the rolling window based on. The DCCB will be triggered once the operation criterion (i.e., for the RHS DCCBs, and for the LHS DCCBs) is satisfied.

5. Case studies

To validate the proposed fault analysis model and the fault detection scheme for hybrid MTDC systems, a ± 200 kV LCC-MMC transmission system illustrated in Fig. 5 is established in PSCAD/EMTDC. The classic constant current control is applied on the 12-pulse LCC-based rectifier, while the droop control for the two MMC-based inverters [25]. Transmission lines are represented by the detailed frequency-dependent model and the classic RL model, respectively. The inductance of smoothing reactors installed at the terminal of LCC is configured as 400mH. The supplementary DC reactors are chosen to be 150mH. Based on and, the threshold values for forward and backward faults can be determined as 51.42 kV and –23.21 kV, respectively. Other concerning parameters of the test system are listed in Table 1.

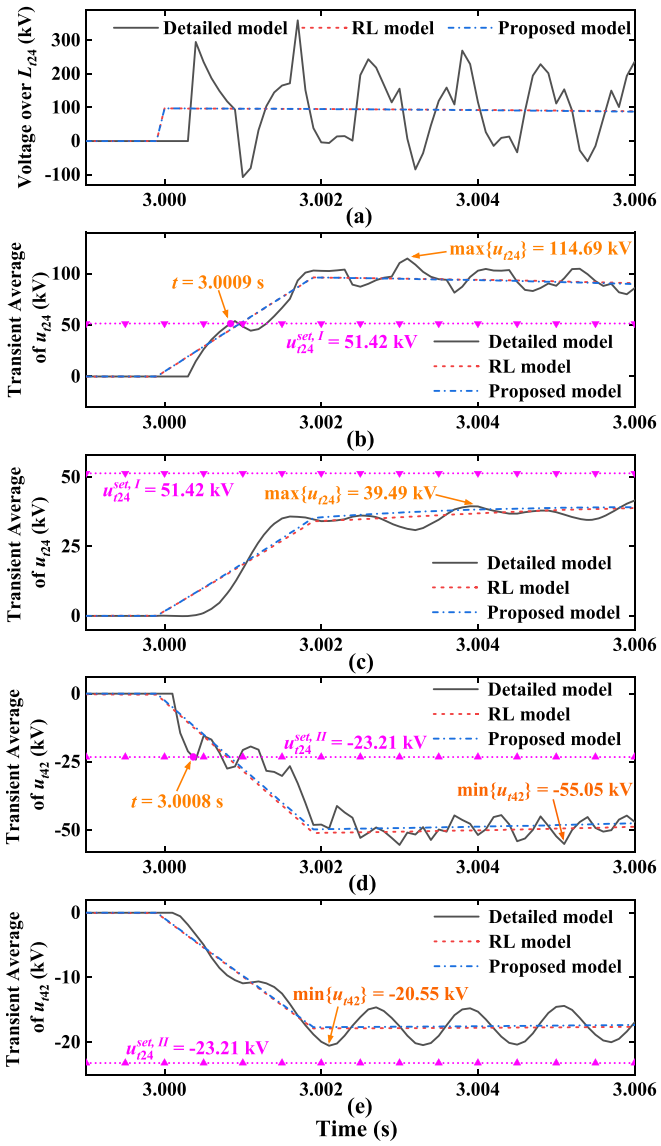


Fig. 10. (a) voltage over L_{t24} when f_2 occurs; (b) transient average of u_{t24} when f_2 occurs; (c) transient average of u_{t24} when f_3 occurs; (d) transient average of u_{t42} when f_1 occurs; (e) transient average of u_{t42} when the severest external backward fault occurs.

5.1. Model verification

Fig. 9 shows the fault component current of the DC grid with the detailed model, the RL model, and the proposed model, respectively. Fig. 9 (a) and (b) are the fault current of Line 14 and Line 24 when a metallic fault occurs at the beginning of Line 14 at $t = 3$ s. It is observed from Fig. 9 (a) that the LCC-connected line current fast increases after DC fault. Subsequently, the fault current reaches its maximum at around 3.004 s and starts to decrease gradually due to the current control effects of LCC. By contrast, the fault current of the MMC-connected line shown in Fig. 9 (b) drastically surges over 6 ms, exhibiting a clear capacitor discharging process. Besides, fault current with the detailed model fluctuates marginally near that with the RL model because of the TW propagation process, but the overall varying trends of fault current with these two models are quite similar. It is indicated from the comparison between the RL model and the proposed model that approximating the nonlinear term based on and neglecting the resistive and capacitive components in the high-frequency region can effectively reduce the model complexity with little loss of accuracy. Fig. 9 (c) and (d) depict the fault current with a metallic fault occurring between L_{t42} and Line

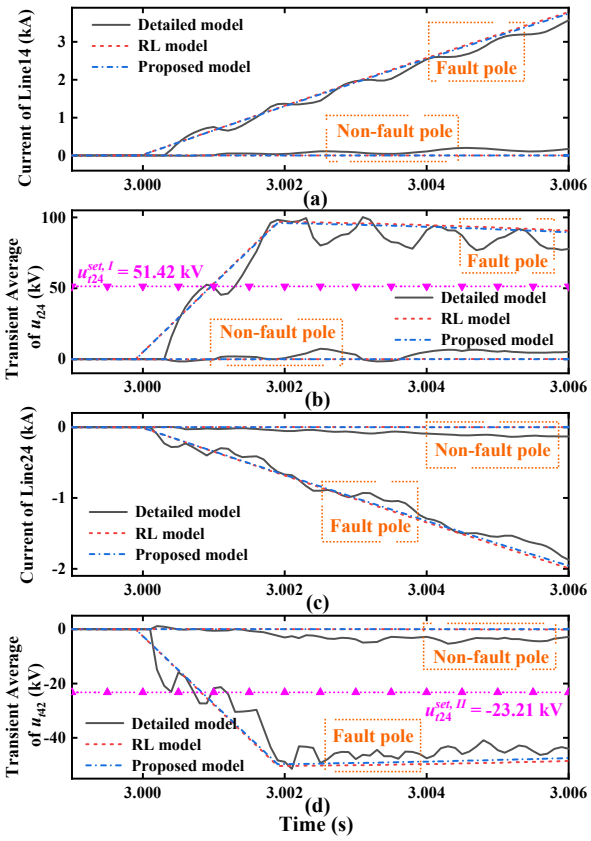


Fig. 11. (a) fault component current of Line 24 when f_2 occurs; (b) transient average of u_{t24} when f_2 occurs; (c) fault component current of Line 24 when f_1 occurs; (d) transient average of u_{t42} when f_1 occurs.

24. As illustrated by Fig. 9 (c), there is a reduction of current of Line 14 and more apparent TW propagations than those shown in Fig. 9 (a), owing to the increase in fault distance. For MMC-connected lines shown in Fig. 9 (d), the fault line current increases apparently faster than that of the healthy line. In addition, the errors between fault current with all the involved models are comparably small, which well manifests the accuracy of the proposed model in short-circuit current calculation within the first instants after fault.

5.2. Fault identification performance

Fig. 10 depicts the dynamics of DC reactor voltage under various fault conditions. Fig. 10 (a) compares the voltage over L_{t24} with three models under a metallic fault occurring between L_{t42} and Line 24 (the remotest internal fault). It is observed that reactor voltage with the detailed model vibrates significantly around those with the RL and the proposed model. That is because TW propagations bring about small fluctuations of DC-link current, while the DC reactor voltage reflects the time derivative of DC-link current, making the fluctuations much sharper. Fortunately, the vibration of reactor voltage can be effectively suppressed by calculating its transient average value, as illustrated by Fig. 10 (b). The transient average of reactor voltage first exceeds the threshold at $t = 3.0009$ s, and then almost stays higher than this value for several milliseconds. This indicates the proposed transient average of reactor voltage is indeed a steadier indicator for DC fault detection. Fig. 10 (c) depicts the transient average of voltage over L_{t24} with the severest external fault that occurs between L_{t43} and Line 34. The maximum transient average voltage with the detailed model is 39.49 kV, which is considerably smaller than the threshold value calculated as 51.42 kV based on. Thus, the protection will not mal-operate under external forward faults. The protection performance with backward

Table 2
Fault identification results under different fault locations and fault types.

Fault line	Fault distance	Fault type	Maximum transient average voltage	Fault identification result	Trigger the breaker?
Line 14	0 km (at the left of L_d)	P2P	−20.55 kV	External backward fault	No
Line 14	0 km (at the left of L_d)	P2G	−18.58 kV	External backward fault	No
Line 14	0 km (between L_d and Line 14)	P2P	−55.05 kV	Internal backward fault	Yes
Line 14	0 km (between L_d and Line 14)	P2G	−51.40 kV	Internal backward fault	Yes
Line 14	25 km	P2P	−66.43 kV	Internal backward fault	Yes
Line 14	25 km	P2G	−63.82 kV	Internal backward fault	Yes
Line 14	50 km	P2P	−80.19 kV	Internal backward fault	Yes
Line 14	50 km	P2G	−76.87 kV	Internal backward fault	Yes
Line 24	0 km	P2P	179.70 kV	Internal forward fault	Yes
Line 24	0 km	P2G	178.25 kV	Internal forward fault	Yes
Line 24	50 km	P2P	139.08 kV	Internal forward fault	Yes
Line 24	50 km	P2G	136.42 kV	Internal forward fault	Yes
Line 24	100 km	P2P	114.69 kV	Internal forward fault	Yes
Line 24	100 km	P2G	100.26 kV	Internal forward fault	Yes
Line 34	0 km	P2P	39.49 kV	External forward fault	No
Line 34	0 km	P2G	38.45 kV	External forward fault	No
Line 34	50 km	P2P	39.07 kV	External forward fault	No
Line 34	50 km	P2G	37.94 kV	External forward fault	No
Line 34	100 km	P2P	38.73 kV	External forward fault	No
Line 34	100 km	P2G	37.58 kV	External forward fault	No

metallic faults is examined in Fig. 10 (d) and (e), which correspond to DC faults happening between L_d and Line 14, and at the left terminal of L_d, respectively. As depicted in Fig. 10 (d), under the remotest internal backward fault, the transient average of reactor voltage fast reduces below the threshold. For the severest external backward fault depicted in Fig. 10 (e), however, the transient average voltage remains higher than the threshold value and thus the protection can prevent mal-operating. The four scenarios mentioned above well validate the selectivity of the fault detection scheme.

Fig. 11 shows the dynamics of fault component current and the transient average of DC reactor voltage under P2G faults. When a metallic P2G fault occurs between L_{t42} and Line 24 (the remotest internal forward fault), the fault component current in the fault pole drastically rises, as depicted in Fig. 11 (a), while the change of the non-fault pole current remains almost constant. Although the non-fault pole current with the detailed model slightly increases due to the coupling effects of the transmission line, the fault current dynamics with the three models are quite similar. Fig. 11 (b) shows the transient average of voltage over L_{t24}. As can be seen, the transient average of the fault pole voltage surges over the threshold immediately following the occurrence of the P2G fault. In contrast, the transient average of the non-fault pole voltage remains far below the threshold. In this situation, the fault pole of the fault line should be cut off while the non-fault pole keeps in

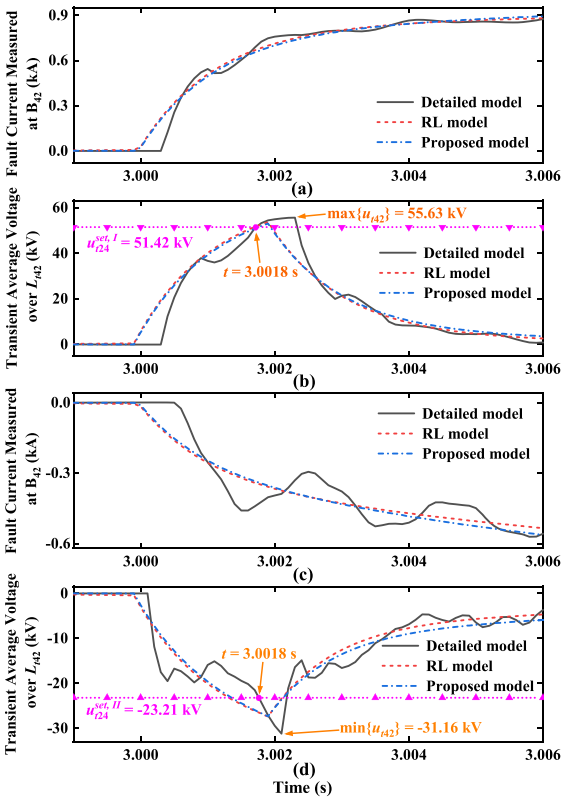


Fig. 12. (a) fault component current measured at B_{42} when f_2 with a fault resistance of 300 Ω occurs; (b) transient average of u_{t42} when f_2 with a fault resistance of 300 Ω occurs; (c) fault component current measured at B_{42} when f_1 with a fault resistance of 300 Ω occurs; (d) transient average of u_{t42} when f_1 with a fault resistance of 300 Ω occurs.

operation. The case of the remotest internal backward fault is shown in Fig. 11 (c) and (d). Similarly, the fault pole current increases fast while the transient average voltage exceeds the threshold value. The changes in electrical quantities in the non-fault pole are rather small so that the protection will not be triggered. Fig. 11 validates the effectiveness of the proposed method under P2G faults.

Table 2 shows the fault identification results of the proposed method under different fault scenarios, which cover internal and external, forward and backward faults, the closest to the remotest fault distance, and different fault types. When the fault occurs at the left of L_d, the maximum transient average voltage is larger than the threshold set as –23.21 kV, no matter whether it is a P2P or P2G fault. As a result, the external backward fault can be recognized correctly, and the DCCB will not be triggered. In contrast, when the fault occurs on Line 14, the protection can identify the internal backward fault as the maximum transient average voltage is lower than the threshold value. Similarly, it is observed from Table 2 that under internal forward fault, the maximum transient average voltage will always exceed the threshold value set as 51.42 kV. As for the external forward fault, the transient average voltage keeps below the threshold so that the protection will not mal-operate. Table 2 validates the effectiveness of the proposed fault detection scheme under various fault locations and fault types.

5.3. *Tolerance capability to fault resistance*

The tolerance capability to fault resistance of the proposed detection scheme is examined in Fig. 12. Fig. 12 (a) and (b) exhibit the fault component of the measured current and the transient average voltage at L_{t42} under the remotest internal forward fault with a fault resistance of 300 Ω, respectively. It is observed in Fig. 12 (a) that there is an apparent saturation trend of the fault current. The reason for this phenomenon is

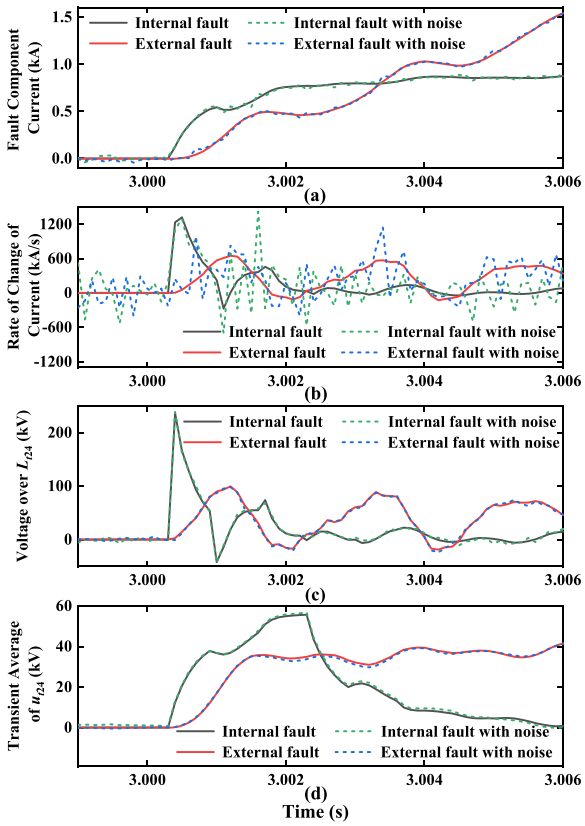


Fig. 13. (a) fault component current measured at B_{24} and with noise; (b) ROCOC of original and corruption signals; (c) voltage over L_{t24} and that with noise; (d) transient average of the original and corrupted DC reactor voltage.

the increase of the total branch resistance reduces the time constant as expressed in, making the fault current quickly approaches the post-fault steady-state value. Moreover, it is indicated the short-circuit current with the proposed model can well approximate that with the detailed model. As illustrated by Fig. 12 (b), the transient average of u_{t42} rises larger than the threshold at $t = 3.0018$ s so that B_{42} can quickly cut off the fault line. Similarly, Fig. 12 (c) and (d) show the case of the remotest internal backward fault with the fault resistance of 300 Ω. It is verified the protection criterion can also be satisfied within 2 ms after the fault, implying the proposed fault detection has relatively high tolerance capability to fault resistances.

5.4. Comparison with the ROCOC-based method

Fig. 13 shows the impacts of noise on the ROCOC-based and the proposed methods. The internal fault is a 300 Ω fault occurring between L_{t42} and Line 24, while the external fault is a metallic fault occurring between L_{t43} and Line 34. The signal-to-noise ratio (SNR) is set as 35 in this test. As shown in Fig. 13 (a), the corrupted fault current denoted by the dashed line is quite close to the original fault current denoted by the solid line, no matter the internal or external fault, indicating the noise corruption is relatively slight. However, it is observed in Fig. 13 (b) that the noise may induce significant distortions in the ROCOC. Particularly, the corrupted ROCOC under the external fault denoted by the dashed blue line reaches almost as high as that under the internal fault denoted by the dashed green line, which implies that the selectivity of the ROCOC-based method can be threatened by noise corruption. Fig. 13 (c) shows the dynamics of the DC reactor voltage, which is very similar to those of the ROCOC without noise. This can be explained by the fact that the DC reactor voltage correlates with the ROCOC as expressed by. As shown in Fig. 13 (d), the transient average of the corrupted DC reactor voltage is almost the same as that of the original voltage, so that the internal and external faults can be accurately distinguished. Fig. 13 manifests that the proposed method is much more robust against noise corruption than the ROCOC-based method.

Conclusion.

The initial DC fault analysis and fault detection scheme of hybrid MTDC systems are investigated in this paper. Using Taylor expansion, LCC with the control effects is equivalent to an RLC circuit for DC fault analysis. By reserving only the high-frequency components of the original Laplace fault circuit, the hybrid MTDC network is significantly reduced at the expense of negligible loss of accuracy. Accordingly, concise analytical expressions of the initial short-circuit current of the fault and healthy lines are derived. On top of this, a fault detection scheme for hybrid MTDC systems based on the transient average of DC reactor voltage is further proposed. The transient average of DC reactor voltage is steadier and more robust with a distinctive focus on fault characteristics of a relatively long time scale. Besides, the thresholds for detecting forward and backward faults can be easily calculated without relying on numerical simulations. Simulation analysis validates the accuracy of the proposed fault analysis model, as well as the capability of fast and reliable fault identification of the transient average voltage-based protection scheme under various fault distances, fault types, and fault resistances.

CRediT authorship contribution statement

Jiapeng Li: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Yujun Li:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Minghao Wang:** Writing – original draft, Writing – review & editing. **Zhao Xu:** Supervision, Writing – original draft, Writing – review & editing. **Guobing Song:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

This project is supported by National Key Research and Development Program of China (2022YFB2402701).

References

[1] Guo C, Zhang Y, Gole AM, et al. Analysis of dual-infeed HVDC with LCC-HVDC and VSC-HVDC. IEEE Trans Power Delivery Jul. 2012;27(3):1529–37.
[2] Zhang F, Xin H, Wu D, et al. Assessing strength of multi-infeed LCC-HVDC systems using generalized short-circuit ratio. IEEE Trans Power Syst Jan. 2019;34(1):467–80.
[3] Tang Y, Li Y. Common lyapunov function based stability analysis of VSC with limits of phase locked loop. IEEE Trans Power Syst 2023;38(2):1759–62. https://doi.org/10.1109/TPWRS.2022.3233762.
[4] Li Y, Tang Y, Lu Y, Hua F, Du Z. Synchronization stability of grid-connected VSC with limits of PLL. IEEE Trans Power Syst 2022. https://doi.org/10.1109/TPWRS.2022.3208244.
[5] Tu Q, Xu Z, Xu L. Reduced switching-frequency modulation and circulating current suppression for modular multilevel converters. IEEE Trans Power Delivery Jul. 2011;26(3):2009–17.
[6] Rao H, Zhou Y, Xu S, et al. Key technologies of ultra-high voltage hybrid LCC-VSC MTDC systems. CSEE J Power Energy Syst Sept. 2019;5(3):365–73.
[7] Li Y, Cao R, Huang W, et al. Design and simulation for the control and protection of hybrid LCC-VSC MTDC system. In: 2019 IEEE 3rd International Electrical and Energy Conference (CIEEC); 2019. p. 281–6.
[8] Mao X, Wen Y, Wu L, et al. Simulation of LCC-MMC HVDC systems using dynamic phasors. IEEE Access 2021;9:122819–28.

[9] Yang J, Fletcher JE, O'Reilly J. Short-circuit and ground fault analyses and location in VSC-based DC network cables. IEEE Trans Ind Electron Oct. 2012;59(10):3827–37.
[10] Beddard A, Sheridan CE, Barnes M, Green TC. Improved accuracy average value models of modular multilevel converters. IEEE Trans Power Delivery Oct. 2016;31(5):2260–9.
[11] Li C, Zhao C, Xu J, et al. A pole-to-pole short-circuit fault current calculation method for DC grids. IEEE Trans Power Syst Nov. 2017;32(6):4943–53.
[12] Li J, Li Y, Xiong at. al. L. DC fault analysis and transient average current based fault detection for radial MTDC system. IEEE Transactions on Power Delivery, Jun 2020;35(3):1310–20.
[13] Li B, Li Y, He J, et al. An improved transient traveling-wave based direction criterion for multi-terminal HVDC grid. IEEE Trans Power Delivery Oct. 2020;35(5):2517–29.
[14] Marvik JI, D'arco S, Sharifabadi K. Protection scheme for multi-terminal radial VSC HVDC system without communication between terminals. In: CIGRE Int. Symp.-Across Borders-HVDC Syst. Mark. Integr., pp. 1-10, May 2015.
[15] Sneath J, Rajapakse AD. Fault detection and interruption in an earthed HVDC grid using ROCOV and hybrid DC breakers. IEEE Trans Power Delivery Jun. 2016;31(3):973–81.
[16] Tong N, Lin X, Li Y, et al. Local measurement-based ultra-high-speed main protection for long distance VSC-MTDC. IEEE Trans Power Delivery Feb. 2019;34(1):353–64.
[17] Li R, Xu L, Yao L. DC fault detection and location in meshed multiterminal HVDC systems based on DC reactor voltage change rate. IEEE Trans Power Delivery June 2017;32(3):1516–26.
[18] Li C, Gole AM, Zhao C. A fast DC fault detection method using DC reactor voltages in HVdc grids. IEEE Trans Power Delivery Oct. 2018;33(5):2254–64.
[19] Li Y, Li J, Xiong L, et al. DC fault detection in meshed MTDC systems based on transient average value of current. IEEE Trans Ind Electron Mar. 2020;67(3):1932–43.
[20] Chen X, Li H, Wang G, et al. A convolution power-based protection scheme for hybrid multiterminal HVDC transmission systems. IEEE Journal of Emerging and Selected Topics in Power Electronics April 2021;9(2):1655–67.
[21] Zhang C, Huang J, Song G, et al. Non-unit ultra-high-speed line protection for multi-terminal hybrid LCC/MMC HVDC system and its application research. IEEE Trans Power Delivery Oct. 2021;36(5):2825–38.
[22] Liang Y, Jiang L, Li H et al. Fault analysis and traveling wave protection based on phase characteristics for hybrid multiterminal HVDC systems. IEEE J Emerg Select Top Power Electronics.
[23] Bucher MK, Franck CM. Contribution of fault current sources in multiterminal HVDC cable networks. IEEE Trans Power Delivery July 2013;28(3):1796–803.
[24] Saad H, Dennetière S, Mahseredjian J, et al. Modular multilevel converter models for electromagnetic transients. IEEE Trans Power Delivery June 2014;29(3):1481–9.
[25] Xiao L, Xu Z, An T, et al. Improved analytical model for the study of steady state performance of droop-controlled VSC-MTDC systems. IEEE Trans Power Syst May 2017;32(3):2083–93.